



Rajiv Gandhi University of Knowledge Technologies

Catering to the Educational Needs of Gifted Rural Youth of Andhra Pradesh
(Established by the Govt. of Andhra Pradesh and recognized as per Section 2(f) of UGC Act, 1956)

Rajiv Knowledge Valley Campus

Machine Learning Day-3

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Kernel function and types

Support Vector Machine

- Support Vector Machine (SVM) is **supervised learning** algorithm which used for classification as well as regression problems
- However, mainly , it is used for **classification problems** in ML
- The goal of SVM algorithm is to create the **best line** or **decision boundary** that can be segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the feature.

ML

ML

ML

Support Vector Machine

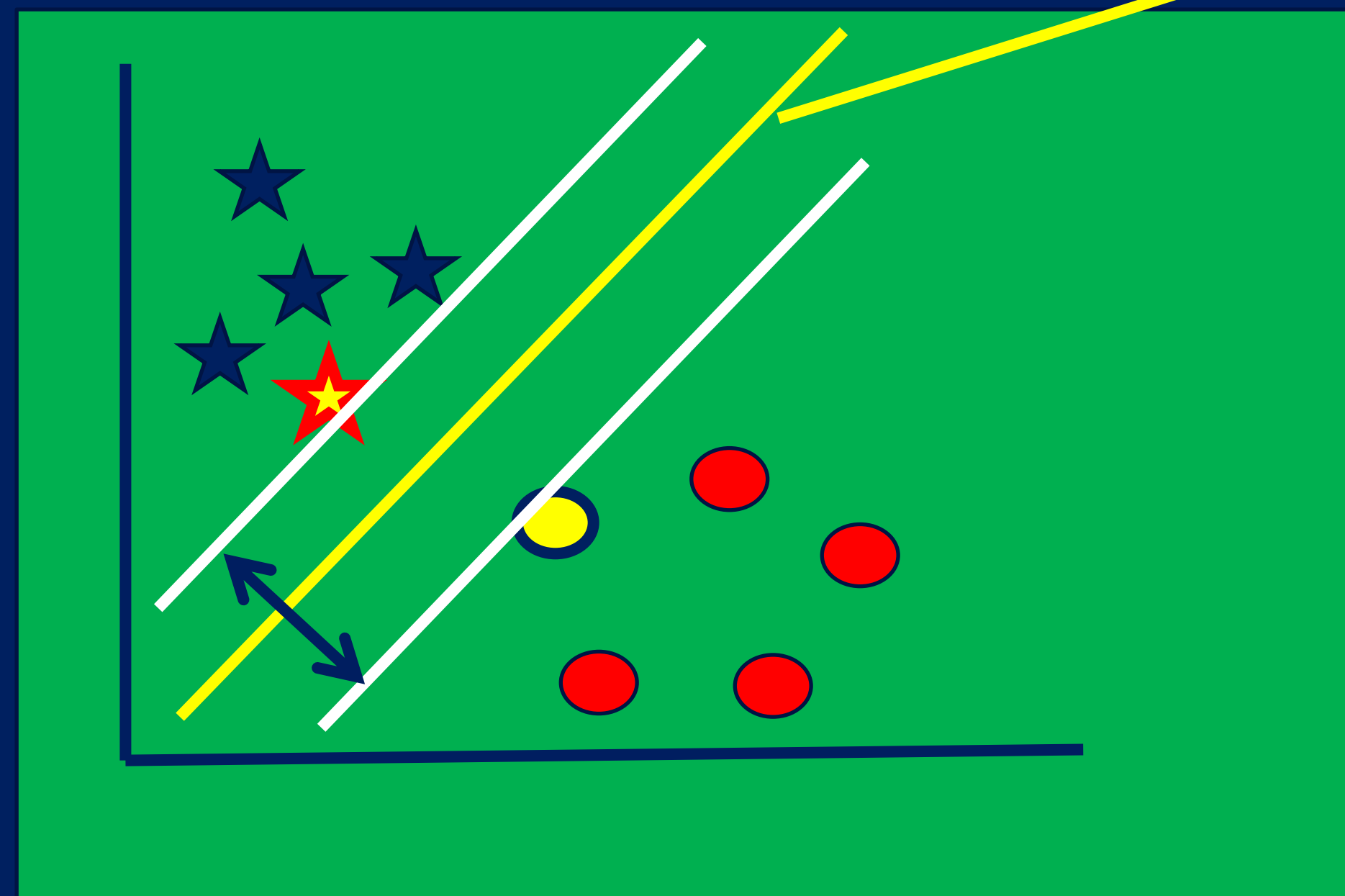
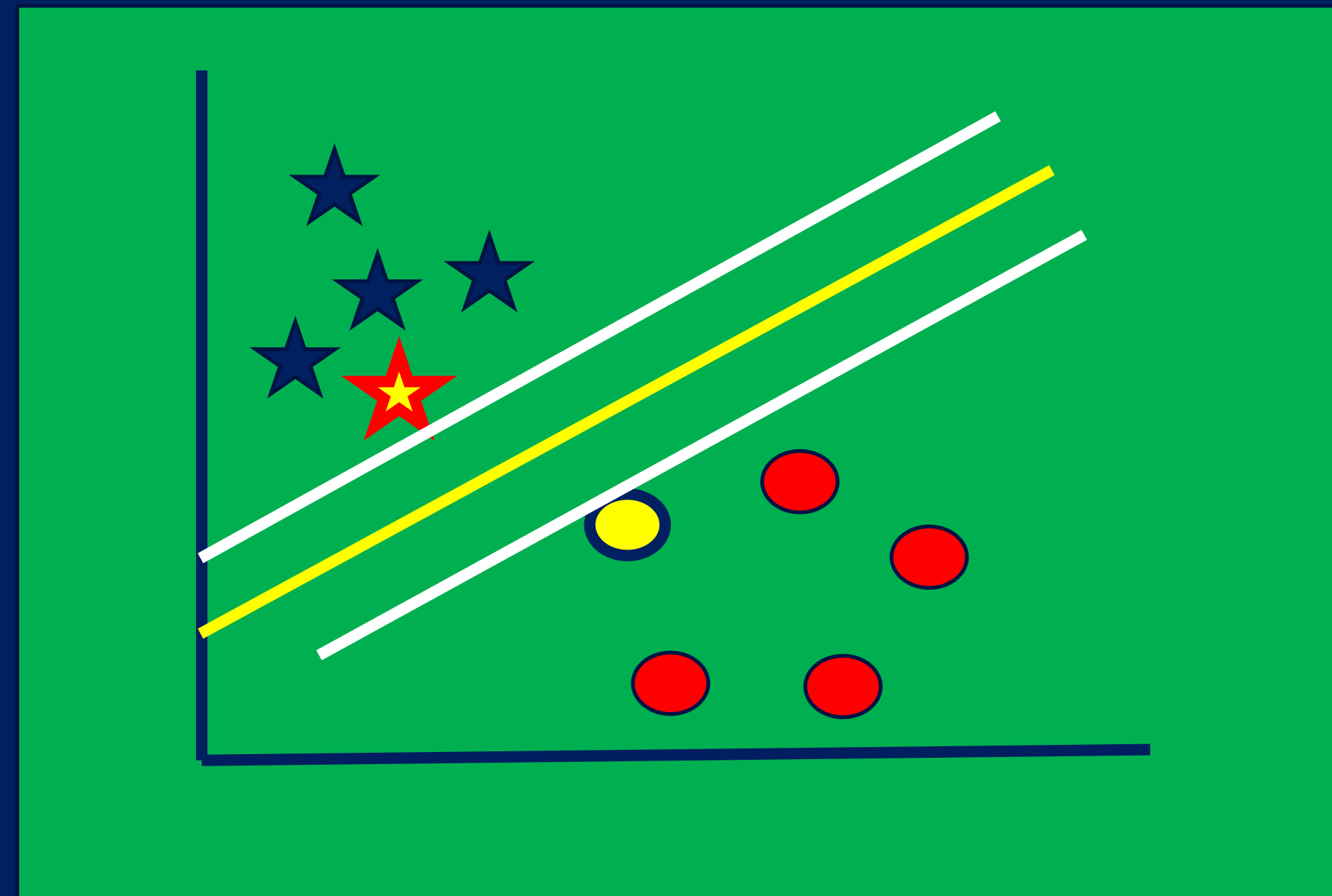
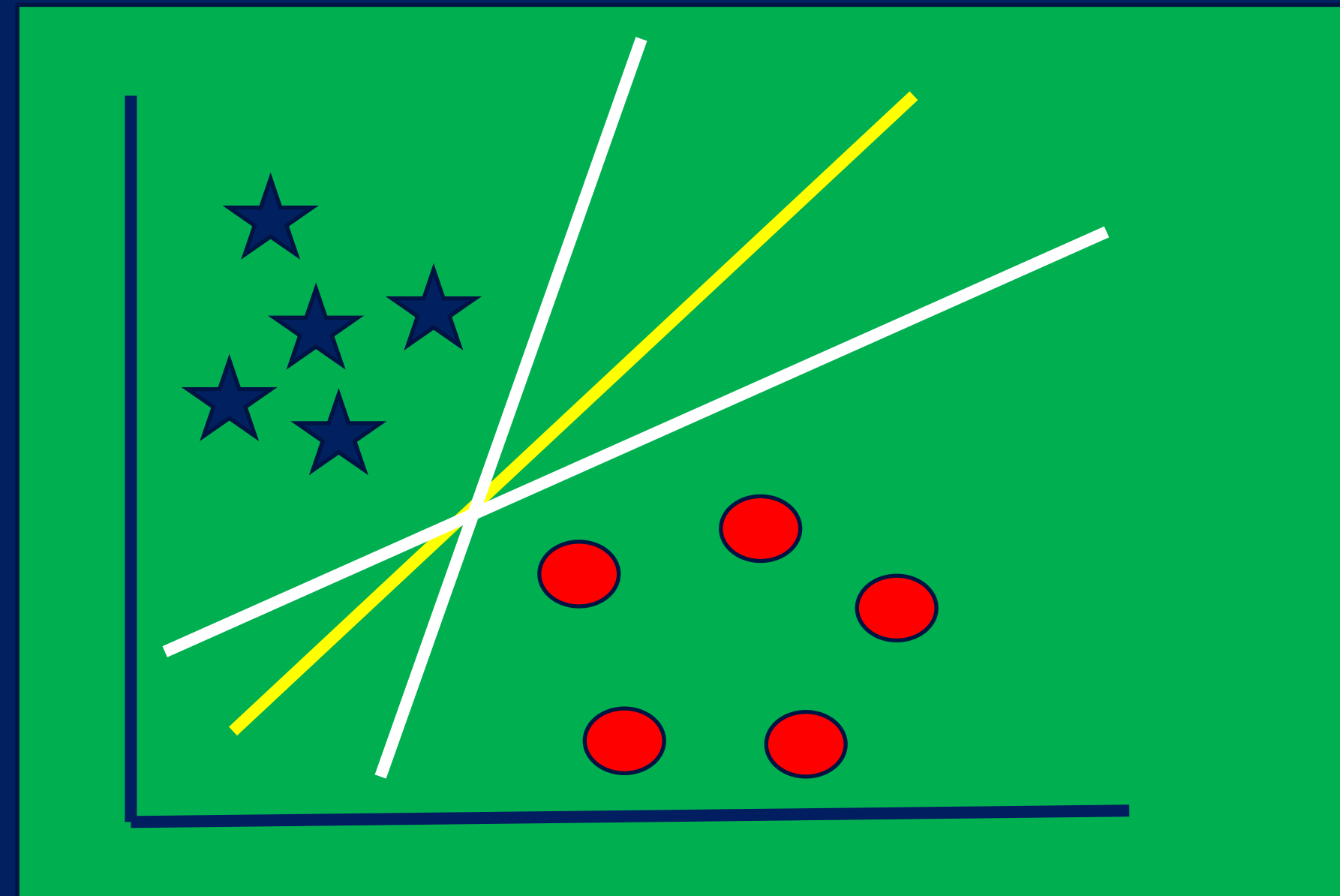
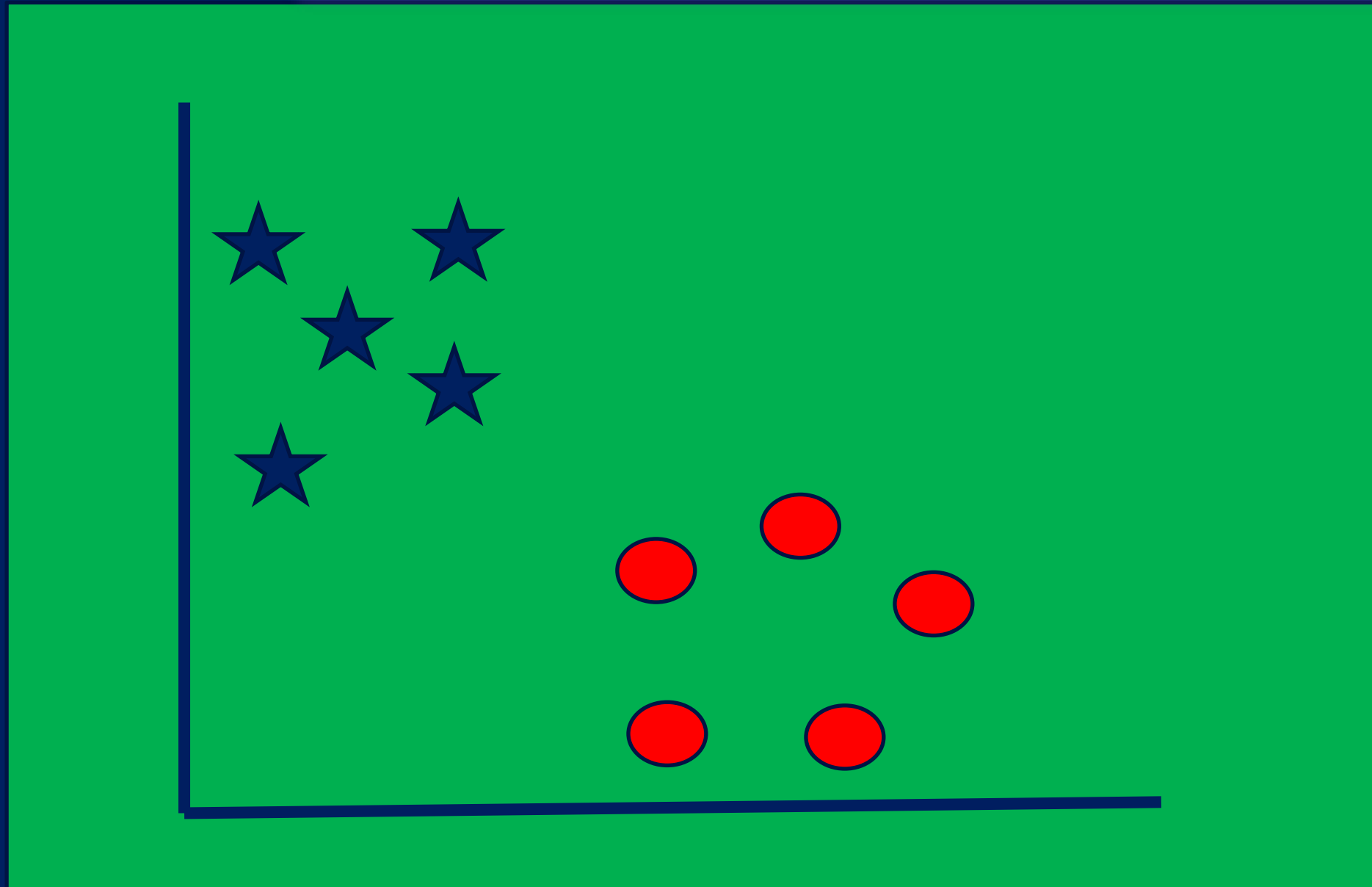
- This best decision boundary is called a “**Hyper plane**”
- We always create a hyper plane that has maximum **margin** which means the maximum distance between support vectors.
- The closest data points to the hyper plane is called “**Support Vectors**”.
- The dimensions of the hyper plane depends on the features present in the dataset which means , if there are two features then hyper plane will be straight line

ML

ML

ML

Support Vector Machine



Hyper plane

Support Vectors

Margin

Types of SVM

Linear SVM

Non - Linear SVM

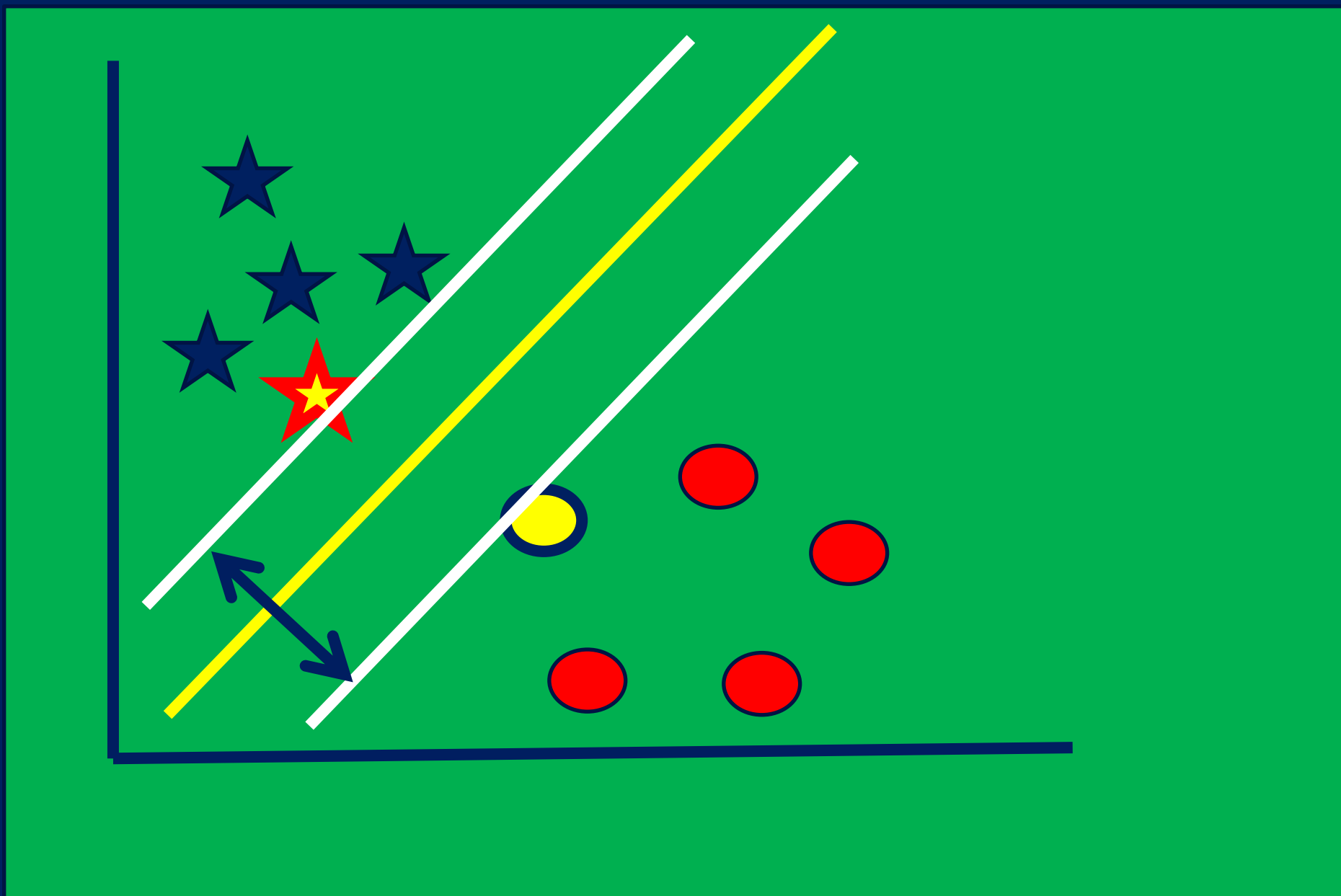
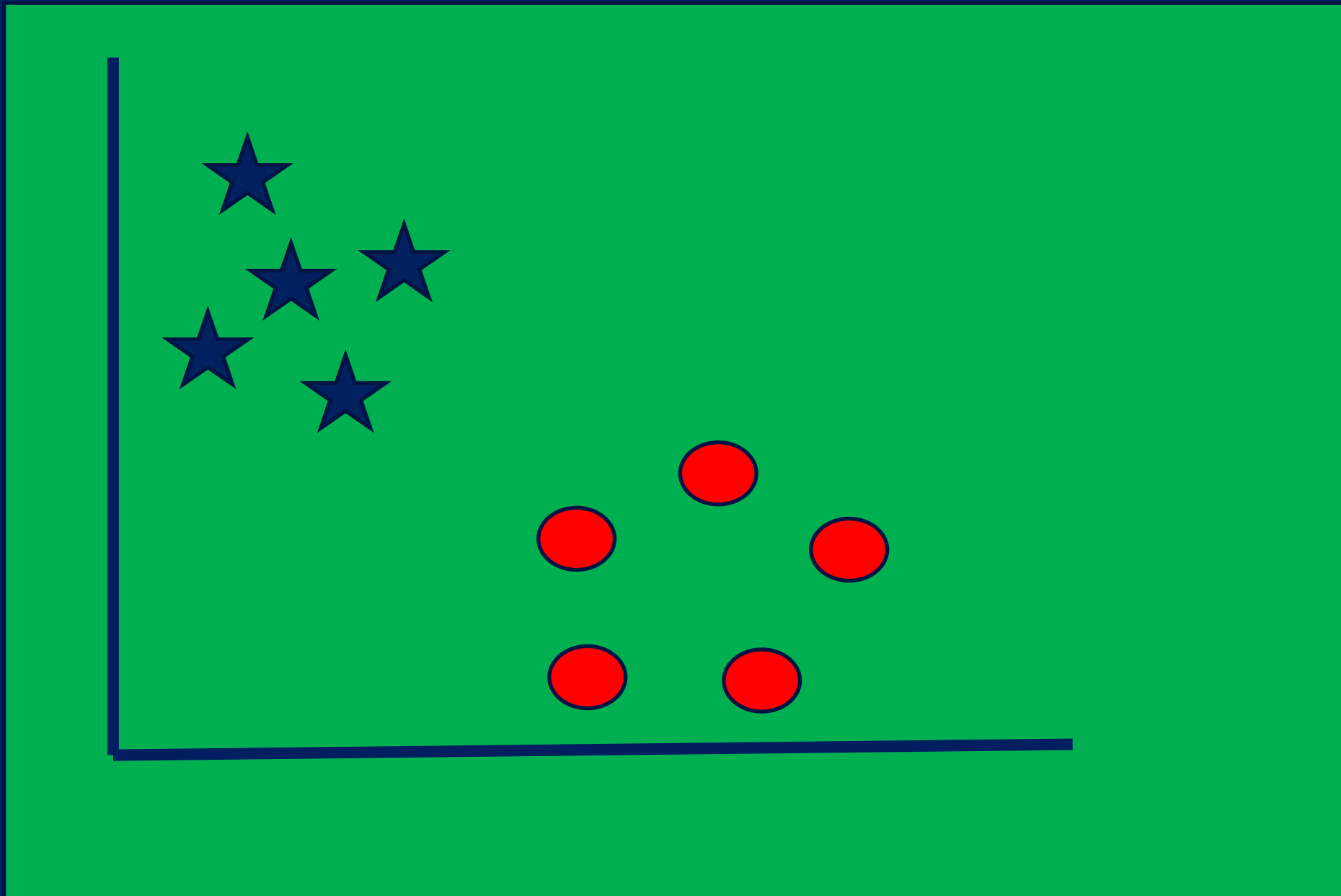
ML

- Linear SVM is used for **linearly separable data**, which means if dataset can be classified into two classes by using a single straight line , then such data is termed as linearly separable data and classifier is used called “**Linear SVM classifier**”
- Non-linear SVM is used for **non-linearly separable data**, which means if dataset cannot be classified by using a straight line , then such data is termed as non-linear data and classifier is used called “**Non-linear SVM classifier**”

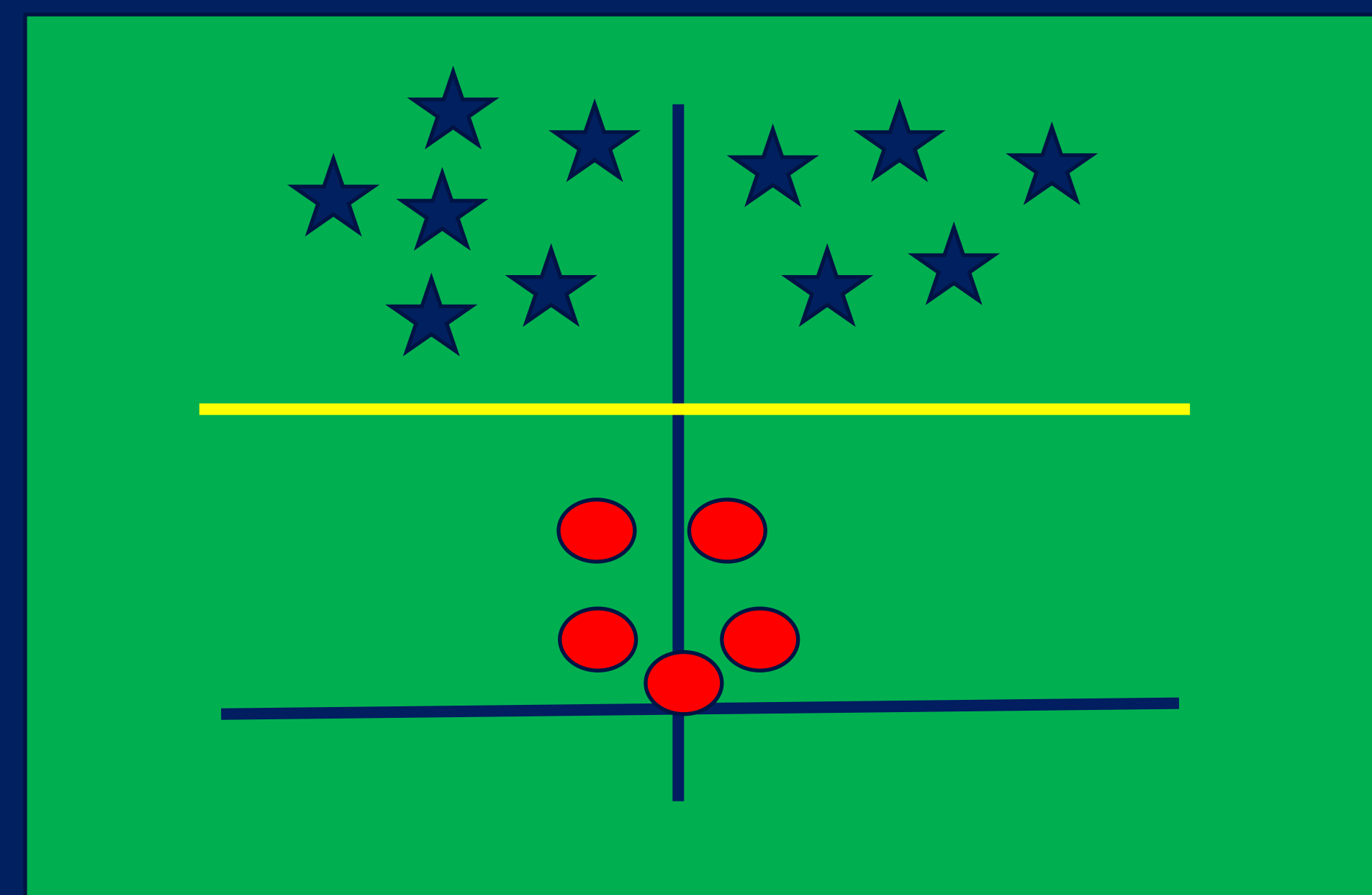
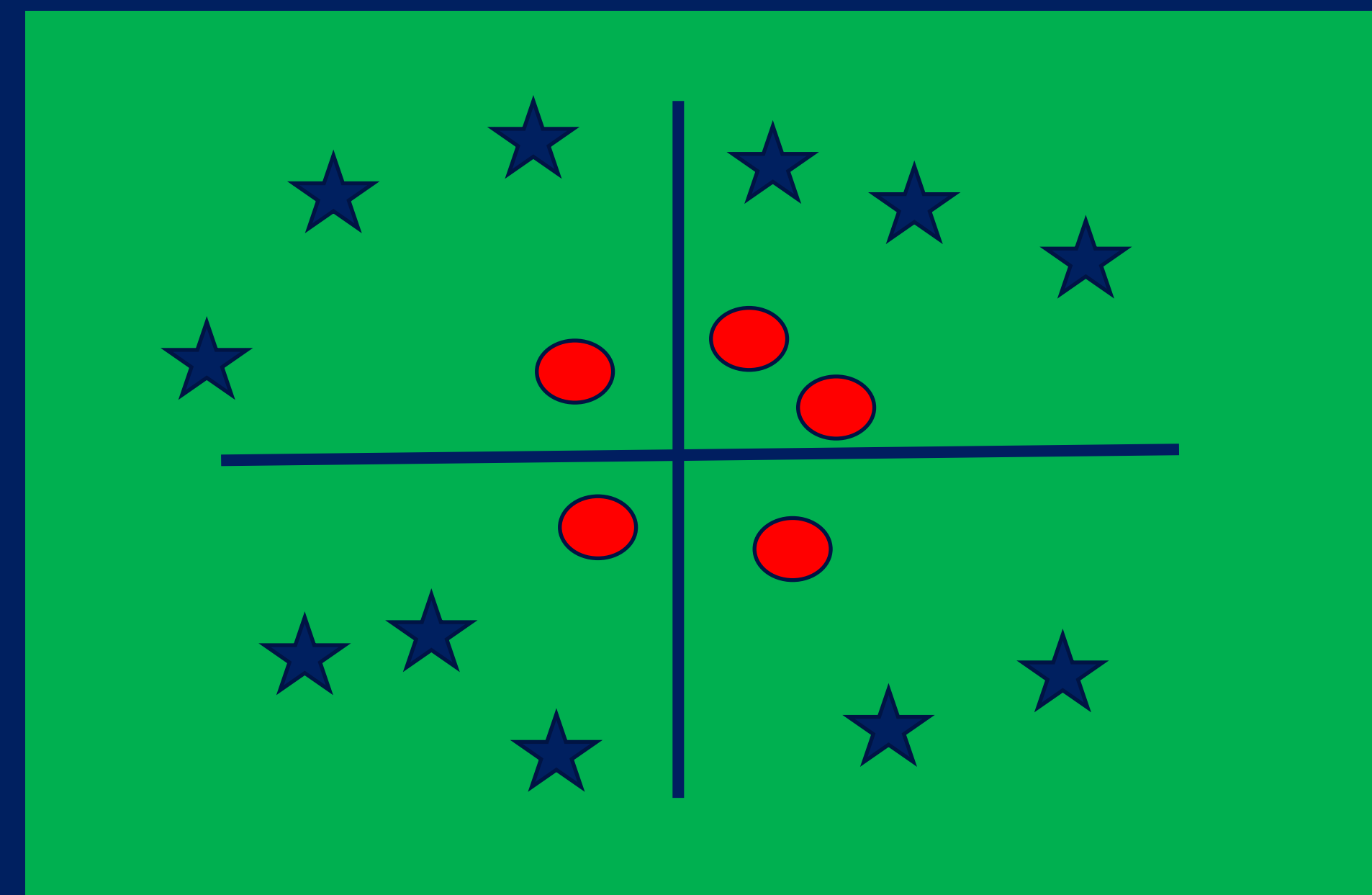
ML

ML

Linear SVM



Non - Linear SVM



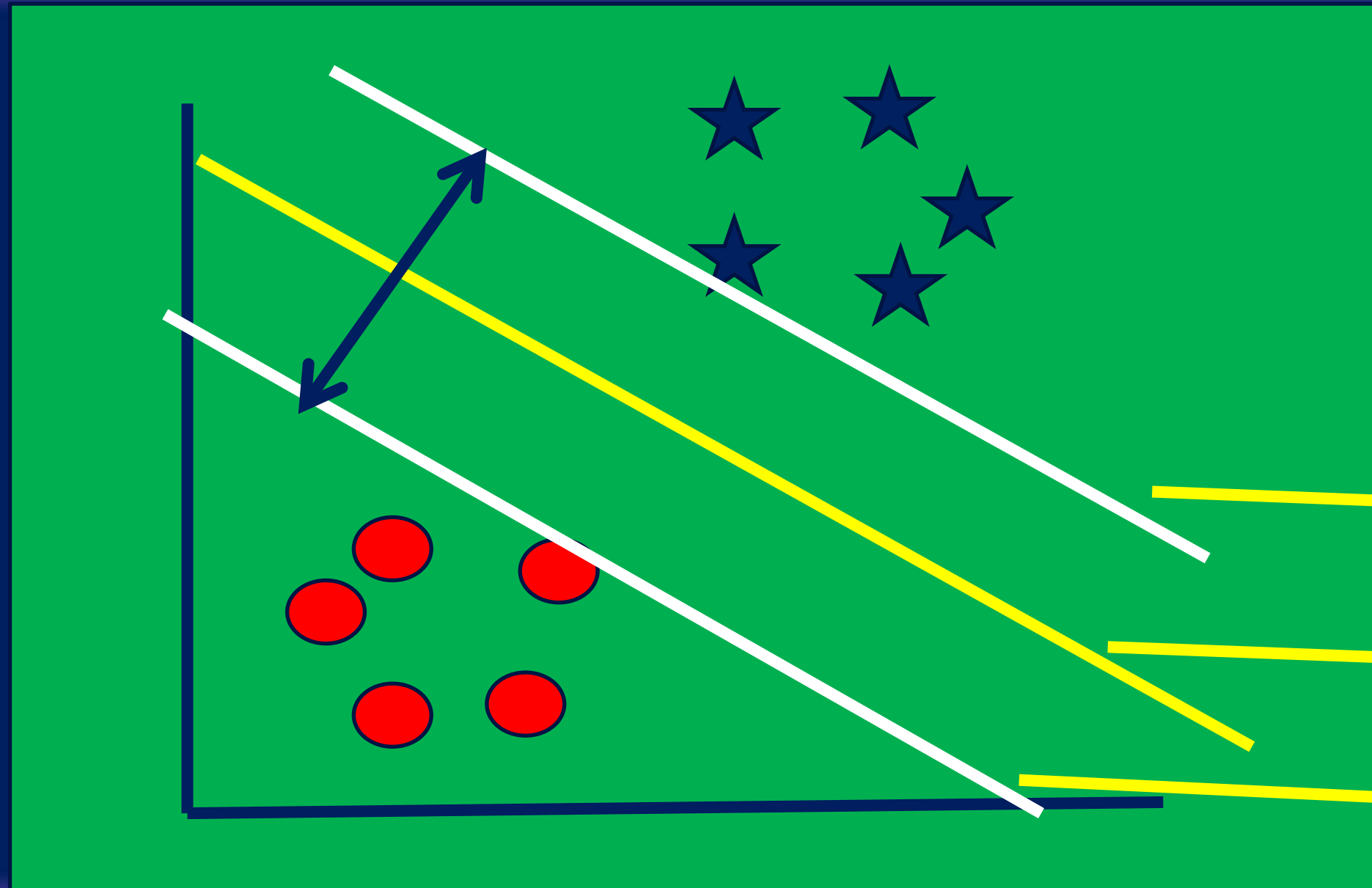
Formulation of SVM

- Assume that we have a two-class training dataset of N points of the form $(\vec{x}_1, y_1), (\vec{x}_2, y_2), \dots, (\vec{x}_N, y_N)$.
where the y_i 's – is output labels. Each $\vec{x}_i$ is a n -dimensional real vector.
- We assume that the dataset is linearly separable.
- Any hyperplane can be written as the set of points $\vec{x} = (x_1, \dots, x_n)$ satisfying an equation of the form
$$\vec{w} \cdot \vec{x} - b = 0$$
- Since the training data is linearly separable, we can select two parallel hyperplanes that separate the two classes of data, so that the distance between them is as large as possible

$$\vec{w} \cdot \vec{x} - b = +1 \quad \text{-----} > 1$$

$$\vec{w} \cdot \vec{x} - b = -1 \quad \text{-----} > 2$$

Formulation of SVM



$$\vec{w} \cdot \vec{x} - b = +1$$

$$\vec{w} \cdot \vec{x} - b = 0$$

$$\vec{w} \cdot \vec{x} - b = -1$$

- For any point on or “above” the hyperplane $\rightarrow 1$, the class label is +1

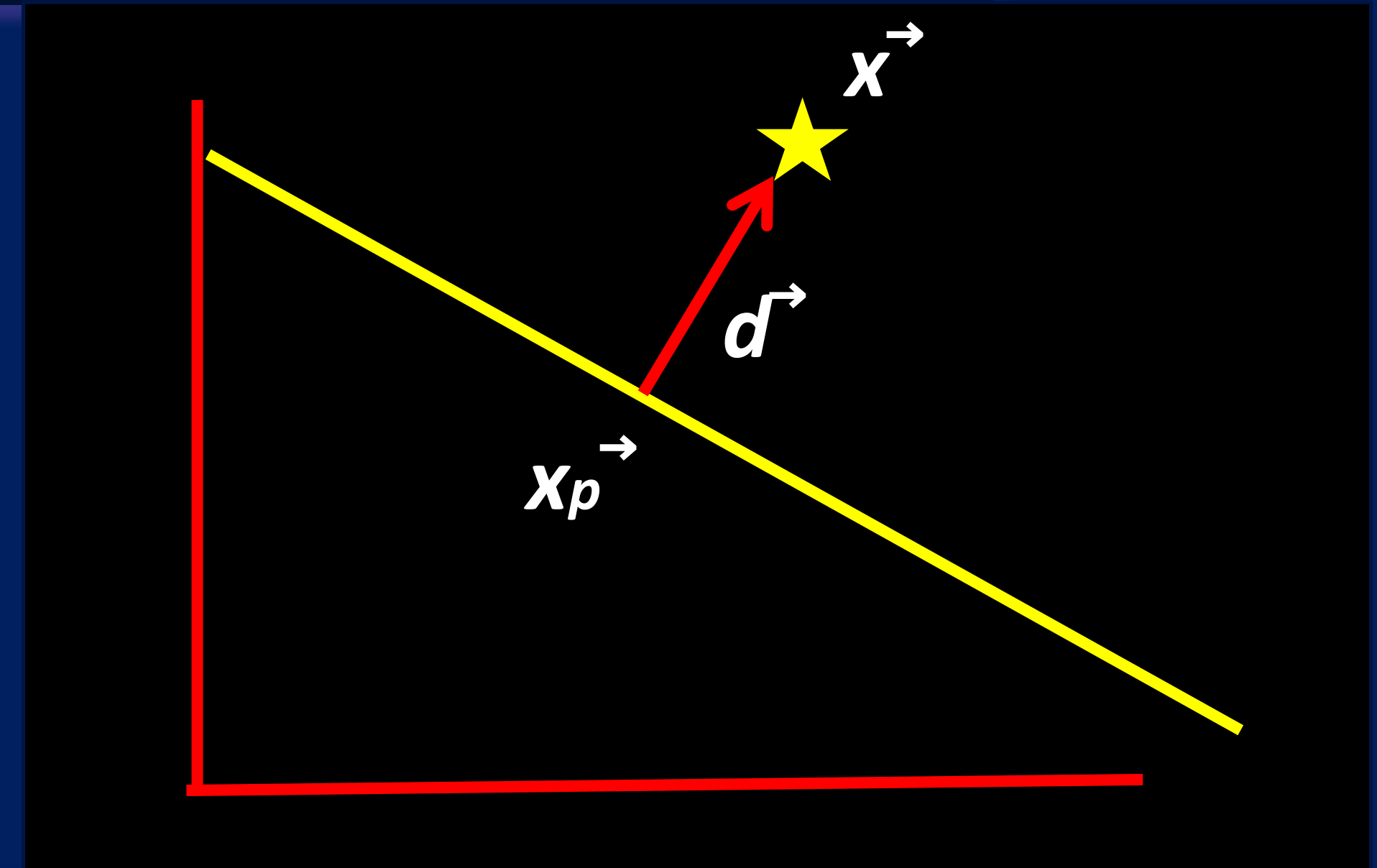
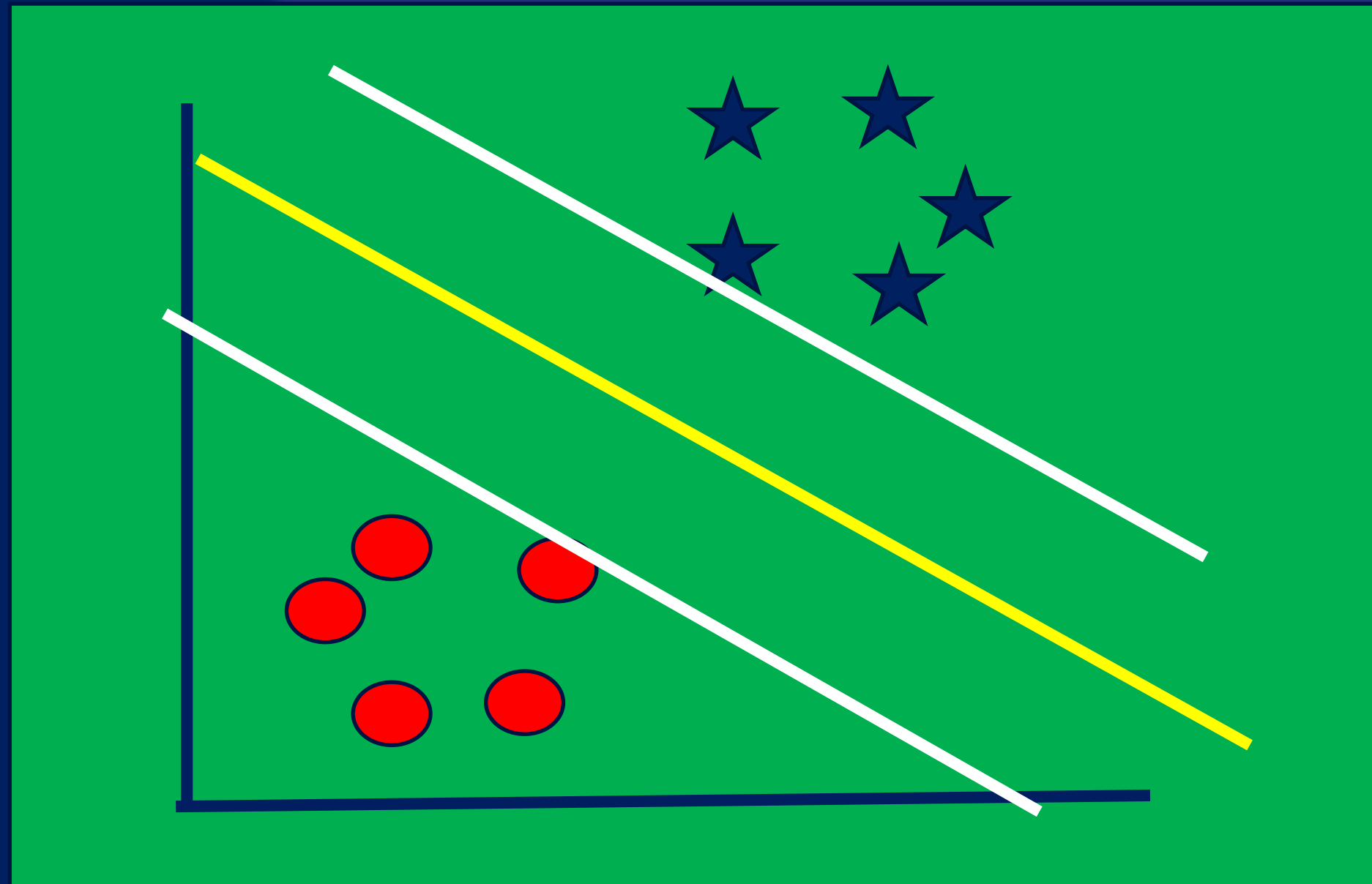
$$\vec{w} \cdot \vec{x}_i - b \geq +1, \quad \text{if } y_i = +1 \quad \text{-----} \rightarrow 3$$

- For any point on or “below” the hyperplane $\rightarrow 2$, the class label is -1

$$\vec{w} \cdot \vec{x}_i - b \leq -1, \quad \text{if } y_i = -1 \quad \text{-----} \rightarrow 4$$

from 3 and 4, we get $y_i(\vec{w} \cdot \vec{x}_i - b) \geq 1, \quad \text{for all } 1 \leq i \leq N$

Formulation of SVM



Now , we find the distance between two hyperplanes.

$$\vec{d} = \alpha \cdot \vec{w}$$

$$\vec{x}_p = \vec{x} - \vec{d}$$

$$\vec{w}^T \cdot \vec{x}_p - b = 0$$

The distance between the two hyperplanes is $2/\|\vec{w}\|$

Formulation of SVM

To maximize the distance between the planes we have to minimize $\|w^{\rightarrow}\|$.

$\|w^{\rightarrow}\|$ is minimum when $1/2 \|w^{\rightarrow}\|_2$ is minimum

Finally SVM is optimization problem

find vectors $w^{\rightarrow}$ and a number b which

minimize $1/2 \|w^{\rightarrow}\|_2$

subject to $y_i(w^{\rightarrow} \cdot x^{\rightarrow}_i - b) \geq 1$, for $i = 1, \dots, N$

Solution

Formulation of SVM

Solution:

The vector $w^{\rightarrow}$ and the scalar b are given by

$$w^{\rightarrow} = \sum_{i=1}^N \alpha_i y_i x_i^{\rightarrow}$$

$$b = 1/2 \left(\min_{i:y_i = +1} (w^{\rightarrow} \cdot x_i^{\rightarrow}) + \max_{i:y_i = -1} (w^{\rightarrow} \cdot x_i^{\rightarrow}) \right)$$

where $\alpha^{\rightarrow} = (\alpha_1, \alpha_2, \dots, \alpha_N)$ is a vector which maximizes

$$\Phi(\alpha_i) = \sum_{i=1}^N \alpha_i - 1/2 \sum_{i=1, j=1}^N \alpha_i \alpha_j y_i y_j (x_i^{\rightarrow} \cdot x_j^{\rightarrow})$$

subject to

$$\sum_{i=1}^N \alpha_i y_i = 0$$

$$\alpha_i > 0 \text{ for } i = 1, 2, \dots, N$$

SVM Classifier Algorithm

1. Given a two-class linearly separable dataset of N points of the form $(\vec{x}_1, y_1), (\vec{x}_2, y_2), \dots, (\vec{x}_N, y_N)$
2. Find $\vec{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_N)$ is a vector which maximizes
$$\Phi(\alpha_i) = \sum \alpha_i - \frac{1}{2} \sum \alpha_i \alpha_j y_i y_j (\vec{x}_i \cdot \vec{x}_j)$$
subject to
$$\sum \alpha_i y_i = 0$$
$$\alpha_i > 0 \text{ for } i = 1, 2, \dots, N$$
3. Compute $\vec{w} = \sum \alpha_i y_i \vec{x}_i$
4. Compute $b = \frac{1}{2} (\min (\vec{w} \cdot \vec{x}_i) + \max (\vec{w} \cdot \vec{x}_i))$
5. Form the SVM classifier function is $f(x) = \vec{w} \cdot \vec{x} - b = 0$,
where α_i is non-zero only if $\vec{x}_i$ is support vector

SVM classifier example

Example1

Using SVM algorithm , find the SVM classifier for the below data

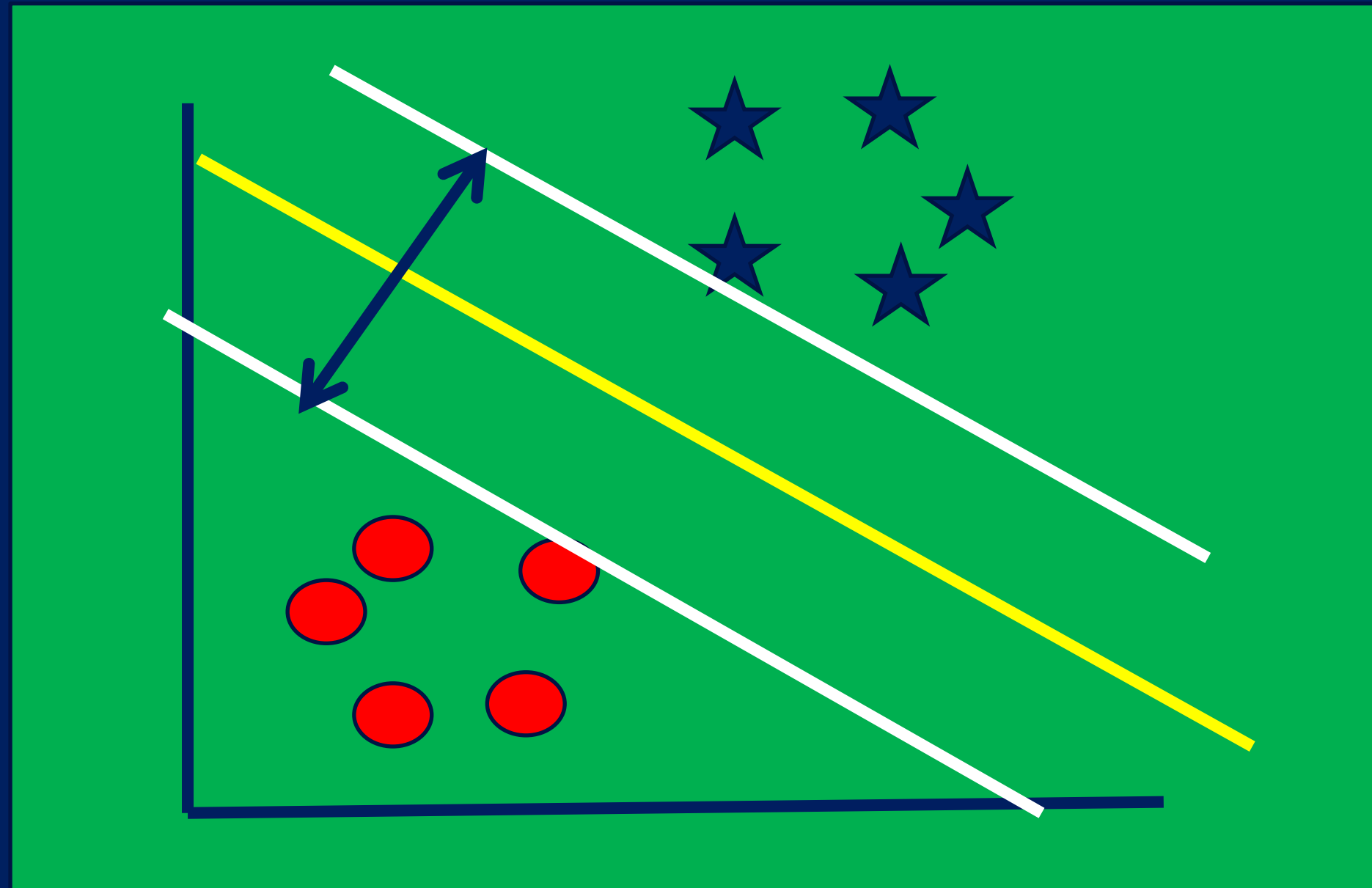
Example	X1	X2	Class
1	2	1	+1
2	4	3	-1

Example2

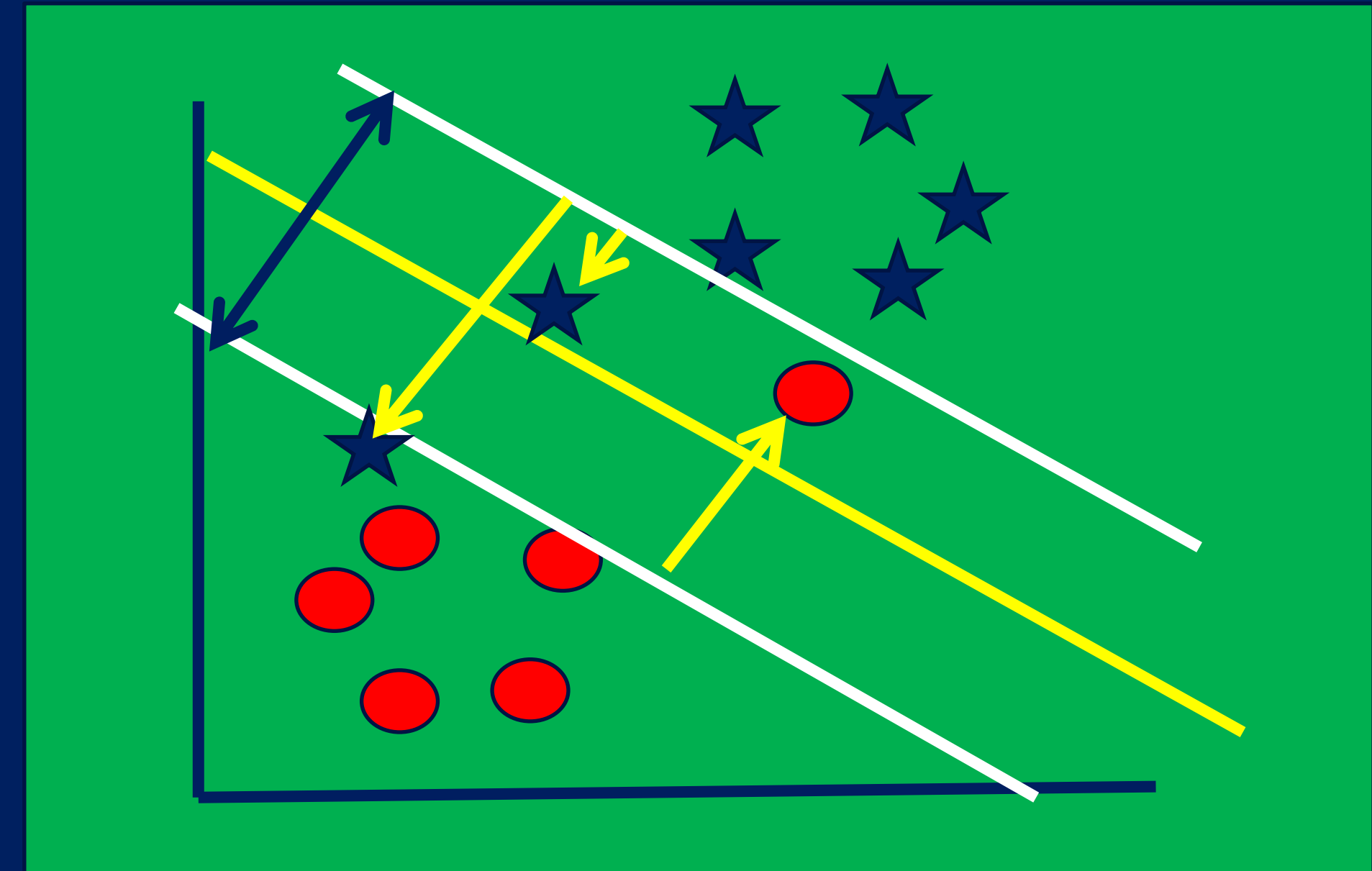
Using SVM algorithm , find the SVM classifier for the below data

Example	X1	X2	Class
1	2	2	-1
2	4	5	+1
3	7	4	+1

Hard Margin Vs Soft Margin



The goal is to find the hyperplane that can perfectly separate the data into two classes without any misclassification



The soft margin is concept that allows for some degree of misclassification or overlap between classes in the training data. To handle this SVM introducing slack variables that allow some data points to be on the wrong side of the margin.

Hard Margin Vs Soft Margin

find vectors $w^{\rightarrow}$ and a number b which
minimize $1/2 \|w^{\rightarrow}\|_2^2$

subject to

$$y_i(w^{\rightarrow} \cdot x^{\rightarrow}_i - b) \geq 1, \quad \text{for } i = 1, \dots, N$$

The hard margin SVM looks for the
widest margin possible

find vectors $w^{\rightarrow}$ and $\xi^{\rightarrow}$ and a number b which
Minimize $1/2 \|w^{\rightarrow}\|^2 + C \sum \xi_i$

subject to $y_i(w^{\rightarrow} \cdot x^{\rightarrow}_i - b) \geq 1 - \xi_i,$

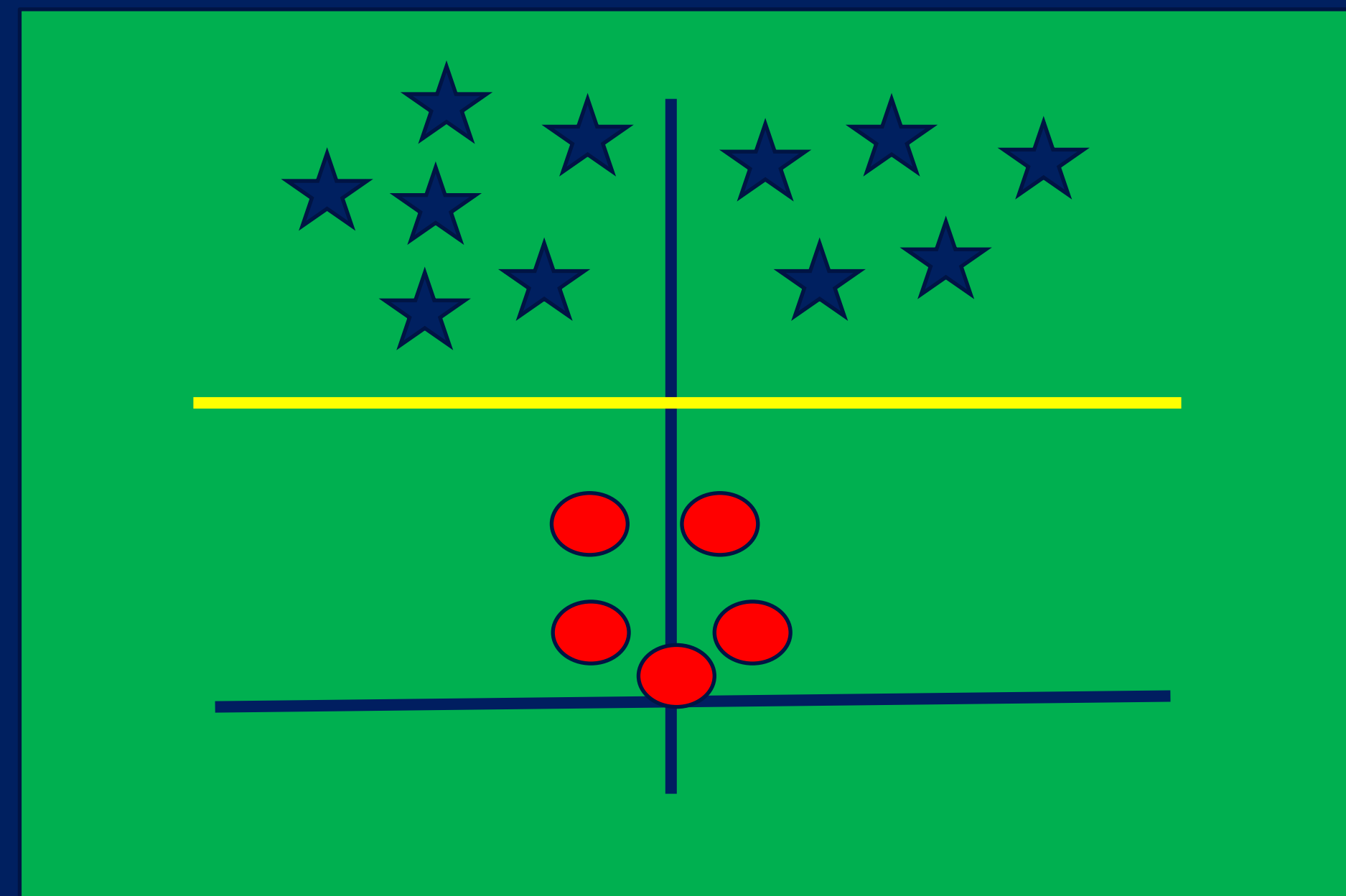
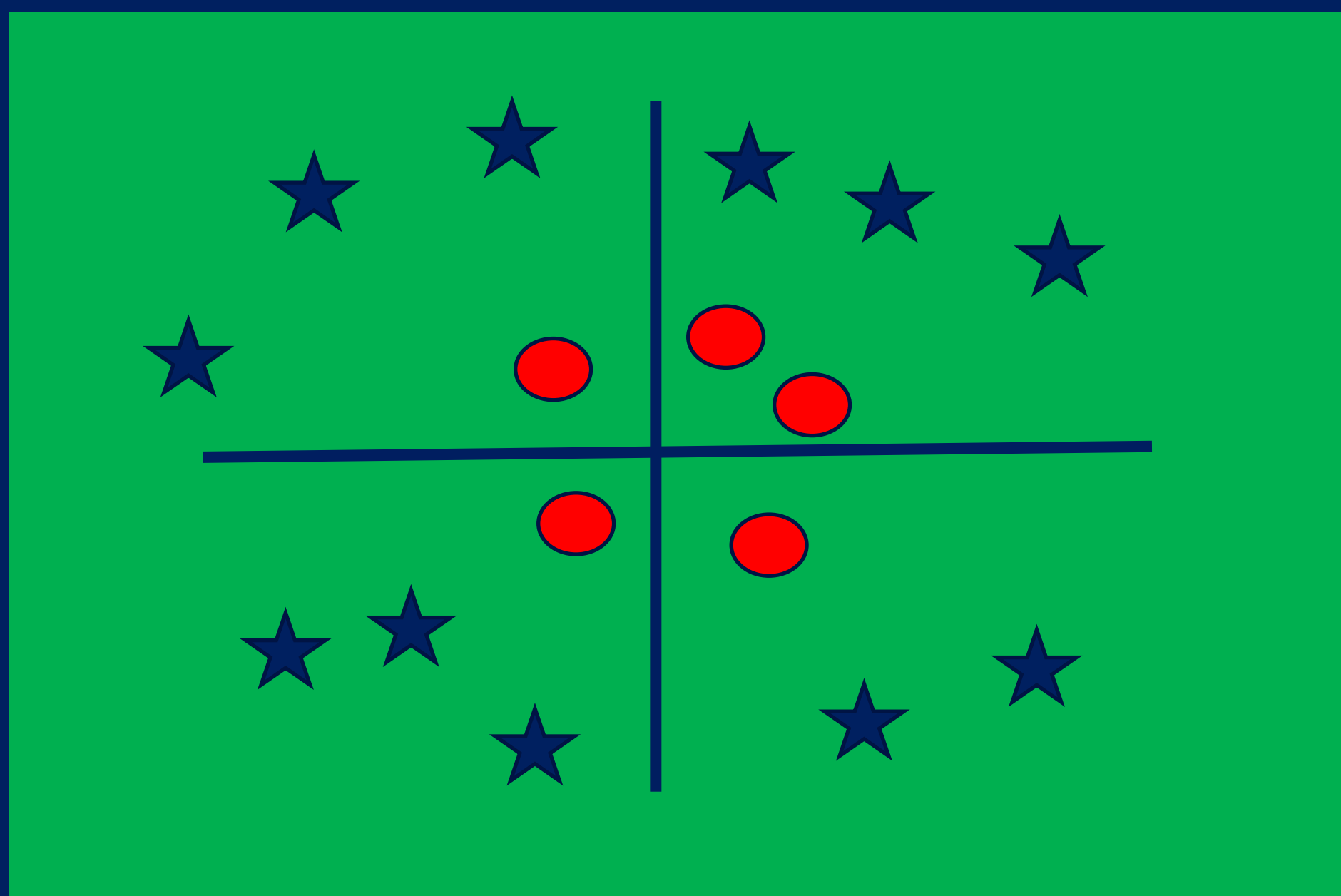
for $i = 1, \dots, N, \xi_i \geq 0, \text{ for } i = 1, \dots, N$

the soft margin SVM finds the hyperplane
that maximizes the margin while minimizing
the penalty for misclassification.

SVM Kernel Function

A kernel function is a function of the form $K(\vec{x}, \vec{y})$, where $\vec{x}$ and $\vec{y}$ are n -dimensional vectors, having a special property. These functions are used to obtain SVM-like classifiers for two-class datasets which are not linearly separable.

Let $\vec{x}$ and $\vec{y}$ be arbitrary vectors in the n -dimensional vector space R^n . Let φ be a mapping from R^n to some vector space. A function $K(\vec{x}, \vec{y})$ is called a kernel function if there is a function φ such that $K(\vec{x}, \vec{y}) = \varphi(\vec{x}) \cdot \varphi(\vec{y})$.



Some of the SVM Kernel Function

we assume that $\vec{x} = (x_1, x_2, \dots, x_n)$ and $\vec{y} = (y_1, y_2, \dots, y_n)$.

1. Homogeneous polynomial kernel

$$K(\vec{x}, \vec{y}) = (\vec{x} \cdot \vec{y})^d$$

where d is some positive integer.

2. Non-homogeneous polynomial kernel

$$K(\vec{x}, \vec{y}) = (\vec{x} \cdot \vec{y} + \vartheta)^d$$

where d is some positive integer and ϑ is a real constant.

3. Radial basis function (RBF) kernel

$$K(\vec{x}, \vec{y}) = e^{-\|\vec{x} - \vec{y}\|^2 / 2\sigma^2}$$

This is also called the Gaussian radial function kernel

4. Laplacian kernel function

$$K(\vec{x}, \vec{y}) = e^{-\|\vec{x} - \vec{y}\| / \sigma}$$

5. Hyperbolic tangent kernel function (Sigmoid kernel function)

$$K(\vec{x}, \vec{y}) = \tanh(\alpha(\vec{x} \cdot \vec{y}) + c)$$

Kernel trick (method)

1. Choose an appropriate kernel function $K(\vec{x}, \vec{y})$.
2. Formulate and solve the optimization problem obtained by replacing each inner product $\vec{x} \cdot \vec{y}$ by $K(\vec{x}, \vec{y})$ in the SVM optimization problem.
3. In the formulation of the classifier function for the SVM problem using the inner products of unclassified data $\vec{z}$ and input vectors $\vec{x}_i$, replace each inner product $\vec{z} \cdot \vec{x}_i$ with $K(\vec{z}, \vec{x}_i)$ to obtain the new classifier function.